

Problem Set 6
Due: Apr. 5, 2021

1. Section 3.1 Problem 2.

Laplace random variable. Let X have the PDF

$$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}$$

where λ is a positive scalar. Verify that f_X satisfies the normalization condition, and evaluate the mean and variance of X .

2. Section 3.2 Problem 7.

Alvin throws darts at a circular target of radius r and is equally likely to hit any point in the target. Let X be the distance of Alvin's hit from the center.

- (a). Find the PDF, the mean and the variance of X .
- (b). The target has an inner circle of radius t . If $X < t$, Alvin gets a score of $S = 1/X$. Otherwise his score is $S = 0$. Find the CDF of S . Is S a continuous random variable?

3. An image is partitioned into two regions, one white and the other black. A reading taken from a randomly chosen point in the white section will be normally distributed with $\mu = 4$ and $\sigma^2 = 4$, whereas one taken from a randomly chosen point in the black region will have a normally distributed reading with parameters $(6, 9)$. A point is randomly chosen on the image and has a reading of 5. If the fraction of the image that is black is α , for what value of α would the probability of making an error be the same, regardless of whether one concluded that the point was in the black region or in the white region?

4. Show that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Hint: $\Gamma(t) = \int_0^\infty e^{-x} x^{t-1} dx$. Make the change of variables and then relate the resulting expression to the normal distribution.